

(y) → yield (IRR)

Case 1
y → Continuous compounding

$$V(y) = \underline{N e^{-y(T-t)}}$$

$$\Delta V \approx \left(\frac{dV}{dy} \right) \Delta y + \frac{1}{2} \left(\frac{d^2 V}{dy^2} \right) (\Delta y)^2$$

$$\rightarrow \frac{dV}{dy} = \underline{N e^{-y(T-t)}} \times \left\{ \underline{-(T-t)} \right\}$$

$$= -V \cdot (T-t)$$

$$\frac{d^2 V}{dy^2} = V \cdot (T-t)^2$$

$$\underline{\Delta V} = -V (T-t) \cdot \Delta y + \frac{1}{2} V \cdot (T-t)^2 (\Delta y)^2$$

(linear)
(quadratic)

$$\underline{\frac{\Delta V}{V}} = - \underline{(T-t)} \cdot \underline{\Delta y} + \frac{1}{2} (T-t)^2 (\Delta y)^2$$

(% change)

Case II . (annual compounding)

$$V(y) = \frac{N}{(1+y)^{T-t}}$$

$$\Delta V \approx -V \cdot \frac{(T-t)}{1+y} \cdot (\Delta y)$$

$$\frac{\Delta V}{V} \approx - \frac{(T-t)}{1+y} \cdot \Delta y$$

$\rightarrow y$

$$V = N \left[(1+y)^{-(T-t)} \right]$$

$$\frac{dV}{dy} = N \left[(T-t) (1+y)^{-(T-t)-1} \right]$$

$$= - \frac{N \cdot (T-t)}{(1+y)^{T-t} (1+y)}$$

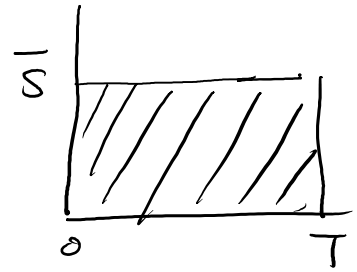
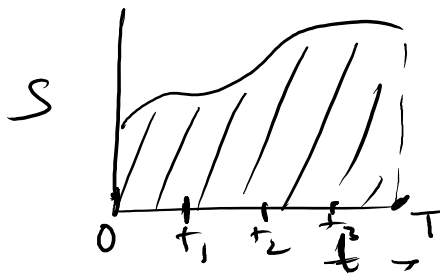
$$= -V \cdot \frac{(T-t)}{(1+y)}$$

Case - III (m times per year compounding)

$$V(y) = \frac{N}{\left(1 + \frac{y}{m}\right)^{m(T-t)}}$$

$$\Delta V \approx - \frac{(T-t)}{1 + y/m} (\Delta y) \quad \leftarrow \frac{0.0001}{\uparrow}$$

$\underline{S(t)}$



$\bar{S}$

$$\bar{S} = \frac{S_1 \cdot t_1 + S_2 \cdot t_2 + S_3 \cdot t_3 + S_4 \cdot t_4}{\sum t_i}$$

$$\bar{S} = \frac{\int_0^T S dt}{\int_0^T dt}$$

$$\bar{X} = \frac{P_1^f x_1 + P_2^f x_2}{(P_1 + P_2) = 1}$$

$$\frac{\int x dp}{\int dp = 1}$$

$$V = \frac{c}{1+y} + \frac{c}{(1+y)^2} + \dots + \dots =$$

$$a + ar + ar^2 + \dots = \frac{a}{1-r}$$

$$V = \frac{c/(1+y)}{1 - \frac{1}{1+y}} = \frac{c/(1+y)}{y/(1+y)} = \frac{c}{y} \rightarrow \Delta V = -\frac{c}{y^2} (\Delta y)$$

$$D = \frac{1 \cdot \left(\frac{c}{1+y}\right) + 2 \cdot \left(\frac{c}{(1+y)^2}\right) + 3 \cdot \frac{c}{(1+y)^3} + \dots}{V}$$

$$S = \frac{1 \cdot \frac{c}{(1+y)} + 2 \cdot \frac{c}{(1+y)^2} + 3 \cdot \frac{c}{(1+y)^3} + \dots}{S \cdot \left(\frac{1}{1+y}\right) = \frac{c}{(1+y)^2} + \frac{2c}{(1+y)^3} + \frac{3c}{(1+y)^4} + \dots}$$

$$S \left(1 - \frac{1}{1+y}\right) = \frac{c}{(1+y)} + \frac{c}{(1+y)^2} + \frac{c}{(1+y)^3} + \dots = V$$

$$\Rightarrow S \cdot \left(\frac{y}{1+y}\right) = V \Rightarrow \frac{S}{V} = \left(\frac{1+y}{y}\right)$$

$$\alpha = \frac{1+y}{y}, \quad \frac{\Delta V}{V} = -\frac{(1+y)/y}{1+y} \cdot (\Delta y) = -\frac{\Delta y}{y}$$

$$\boxed{\Delta V = -\frac{c}{y^2} (\Delta y)}$$



Simply compounded
forward rate $(t, T_1, T_2) = \frac{1}{T_2 - T_1} \left(\frac{P(t, T_1)}{P(t, T_2)} - 1 \right)$

